Improving Class-Conditional GAN Generation with Auxiliary Classification: An Empirical Study of Hard CIFAR-10 Classes

Deep Learning Project, Part C (Individual Challenge Task)
Topic Area: Generative Adversarial Networks (GAN)

Abstract

Standard conditional Deep Convolutional GANs (cDCGANs) generate class-conditioned images by concatenating a label embedding with the generator/discriminator inputs, but this design gives the discriminator no explicit incentive to enforce that generated images match their requested label beyond the implicit adversarial signal. This paper investigates whether adding an auxiliary classification objective, specifically the Auxiliary Classifier GAN (ACGAN) design proposed by Odena et al. [3], improves class-conditional fidelity on CIFAR-10 and whether this comes at a cost to overall image quality, particularly for visually similar classes. Building on a companion study (Part A of this assignment) that identified cat, dog, bird, and horse as the hardest CIFAR-10 classes for a conditional DCGAN, a baseline cDCGAN and an ACGAN were trained under matched conditions (80 epochs, identical generator architecture, identical hyperparameters) and evaluated with Fréchet Inception Distance (FID), Kernel Inception Distance (KID), and an independent classifier's class-conditioning accuracy. The ACGAN raised class-conditioning accuracy substantially, from 63.15% to 71.70% (+8.55 percentage points), but slightly worsened overall distributional quality (FID 258.91 → 267.15; KID 0.0868 → 0.0948), and its hard-class-group mean FID was essentially unchanged (543.62 → 548.50) — with dog improving markedly (885.76 → 784.96 FID) while bird and cat worsened (466.49 → 529.27 and 427.79 → 481.84, respectively). This asymmetric, non-uniform pattern is consistent with the diversity/fidelity trade-off documented in the ACGAN critique literature [14], [15]. A subsequent exponential moving average (EMA) continuation produced the strongest observed balance of distributional quality and class-conditioning accuracy, with the EMA-averaged ACGAN generator recording the best value on every tracked metric (FID 248.78, KID 0.0799, class-conditioning accuracy 76.25%, hard-class mean FID 491.81), ahead of both the plain ACGAN and the baseline. However, because this continuation also included 20 additional fine-tuning epochs at a lower learning rate, the improvement cannot be attributed to EMA alone. The results show that an auxiliary classification loss is an effective, low-complexity route to better class-conditioning, but that its side effects on fidelity are real and were only offset after a further stabilisation stage.

1. Introduction

1.1 Motivation

Generative Adversarial Networks (GANs) [16] learn to synthesise realistic data by pitting a generator against a discriminator in a minimax game. When a class label is available, as in CIFAR-10, the natural extension is a conditional GAN (cGAN) that generates samples belonging to a requested class [2]. The most direct way to build a cGAN, and the approach used for the baseline model in this assignment (and in Part A of this project), is to concatenate an embedding of the label with the generator's noise input and with the discriminator's image input. This design was introduced alongside DCGAN-style convolutional architectures [1], [2].This label-concatenation approach conditions generation implicitly: the discriminator only ever must decide "is this a real image consistent with this label", and nothing forces the generated image's visual class identity to be strongly separable from other classes.

Part A of this assignment trained a series of conditional DCGAN variants on CIFAR-10 and, through a sequence of controlled ablations (latent dimension, batch size, two-time-scale update rule-style learning rates, upsampling depth, self-attention, and model capacity), found that increasing generator/discriminator capacity ("wide/deep") was the single most effective architectural change, and that a properly tracked exponential moving average (EMA) of the Generator weights produced the strongest final result. The final Part A model (wide/deep cDCGAN + EMA, 150 epochs) achieved an FID of 103.86, a KID of 0.0024, a recall of 0.482, a coverage of 0.956, a classifier-based Inception Score of 5.83 ± 0.35, and a class-conditioning accuracy of 80.1%, compared with an early baseline of FID 170.29, KID 0.0893, recall 0.389, Inception Score 4.74 ± 0.34, and class-conditioning accuracy 64.4%. Critically, per-class diagnostics in Part A showed that the improvement was not uniform across classes. Cat, dog, bird and horse remained disproportionately difficult because these visually overlapping animal classes share fur textures, four-legged silhouettes and similar backgrounds. Even in the final Part A model, dog remained the most difficult class, with a per-class FID of 630.51, substantially higher than the other classes.

Building on the architectural and optimisation improvements achieved in Part A, this study investigates a remaining question: does the conditioning mechanism itself, beyond model capacity and training stabilisation, limit how effectively a GAN can be steered towards a requested class, especially for visually overlapping classes? This motivates the investigation of an alternative conditioning strategy in Part C.

1.2 Research Question and Hypotheses

This paper investigates the following research question, adopted directly as the guiding question for the executed experiment:

"Can an auxiliary classification objective improve class-conditional fidelity for visually overlapping CIFAR-10 classes compared with a standard label-conditioned DCGAN?"

Two hypotheses follow directly from the ACGAN literature [3] and from Part A's per-class findings:

- **H1 (conditioning accuracy):** Adding an auxiliary classifier head to the discriminator, which is trained to predict the class of both real and generated images and back-propagates a classification loss into the generator, will increase the accuracy with which generated images match their requested label, relative to label-concatenation conditioning alone.
- **H2 (fidelity trade-off):** Because the auxiliary classifier rewards separability between classes rather than realism per se, gains in conditioning accuracy may come with a cost to overall distributional realism (FID/KID), and the effect may be uneven across the four hard classes identified in Part A (cat, dog, bird, horse) rather than uniformly positive which is a pattern documented for ACGAN in the literature [14], [15].

The study also asks an exploratory, non-preregistered question: whether a lightweight EMA fine-tuning stage, based on the technique that produced the largest single improvement in Part A, can mitigate any fidelity trade-off introduced by the auxiliary classifier by combining ACGAN's conditioning benefit with EMA's stabilisation benefit.

1.3 Contribution and Scope

This paper presents a controlled empirical comparison between a baseline cDCGAN and an ACGAN rather than proposing a new architecture. The generator architecture, hyperparameters, 80-epoch training budget, random seed and evaluation protocol were kept identical. Only the discriminator's conditioning mechanism and loss function were changed. This isolates the effect of the auxiliary classification objective from differences in model capacity or training budget. All reported metrics and figures were obtained directly from the accompanying executed Jupyter notebook.

2. Related Works

2.1 Conditional Image Generation with GANs

Radford et al. [1] introduced DCGAN, a set of architectural conventions (strided/transposed convolutions, batch normalisation, LeakyReLU) that made convolutional GANs practical to train, and which underlies the generator and discriminator backbones used in this study. Mirza and Osindero [2] extended the original GAN framework to the conditional setting by feeding a one-hot (or embedded) class label alongside the noise vector to the generator and alongside the image to the discriminator — the label-concatenation strategy this paper calls the "baseline cDCGAN". This is a simple, general-purpose conditioning mechanism, but, as Mirza and Osindero note, it relies entirely on the adversarial signal to enforce label consistency; there is no explicit term in the loss that rewards or penalises class correctness.

2.2 Auxiliary Classifier GANs and the Diversity/Fidelity Trade-off

Odena, Olah, and Shlens [3] proposed the Auxiliary Classifier GAN (ACGAN), the architecture directly implemented in this paper's experimental model. ACGAN removes the label input from the discriminator and instead adds a second output head that predicts the class of an image (real or fake) in addition to the usual real/fake decision. The generator is then trained not only to fool the discriminator but also to produce images the auxiliary classifier assigns to the requested class. Odena et al. showed that this substantially improves the perceptual quality and class-discriminability of generated ImageNet samples relative to a purely adversarial conditional baseline.

However, follow-up work has identified a systematic weakness in this design. Gong et al. [14] show theoretically that the ACGAN objective, as originally formulated, implicitly maximises the mutual information between generated images and labels in a way that pushes the generator toward low intra-class diversity: when class distributions overlap in the underlying data (exactly the

situation with CIFAR-10's animal classes), ACGAN's classifier term encourages the generator to produce images that are easy to classify rather than images that are faithful to the true conditional distribution, and they propose "Twin Auxiliary Classifiers GAN" (TAC-GAN) with an additional mutual-information estimator to counteract this. Hou et al. [15] argue that the auxiliary classification objective may provide poorly calibrated gradients as the generated distribution changes during training, and propose "Conditional GANs with Auxiliary Discriminative Classifier" (ADC-GAN), which trains the classifier on both real and generated data jointly. Both papers frame the same underlying phenomenon this paper investigates empirically: an auxiliary classifier can inflate apparent class-conditioning accuracy while degrading realism or diversity, particularly for classes whose real-data supports overlap. This paper's Discussion section (Section 5) interprets its own bird/cat regressions directly in light of this literature.

An alternative conditioning strategy that avoids adding a classifier altogether is the projection discriminator of Miyato and Koyama [12], which injects the label by taking an inner product between a learned label embedding and the discriminator's image feature vector, rather than by concatenation or by auxiliary classification. Because projection discriminators do not impose an explicit classification objective, they are reported to avoid the ACGAN diversity collapse while still improving class-conditional fidelity, at the cost of a more involved discriminator architecture. This paper does not implement a projection discriminator, but the comparison motivates why ACGAN's known weaknesses (rather than being unique to poor implementation) are an active area of GAN research and a reasonable, literature-grounded prediction for what this paper's results might show.

2.3 Training Stabilisation Techniques

Several techniques exist to stabilise adversarial training independent of the conditioning mechanism. Miyato et al. [13] introduce spectral normalisation, which constrains the Lipschitz constant of each discriminator layer by rescaling weight matrices by their largest singular value, and which is frequently paired with both projection discriminators [12] and self-attention GANs [11]. Zhang et al. [11] add a self-attention module to both generator and discriminator so that distant spatial regions of an image can be modelled jointly rather than only through local convolutions, and apply spectral normalisation to the generator as well as the discriminator to improve conditioning. Neither spectral normalisation nor self-attention is used in this paper's models. Both the baseline and the ACGAN use conventional strided/transposed convolutions without these additions, but they are noted here as established, complementary techniques that could plausibly extend this work.

The specific stabilisation technique used in this paper, on top of the ACGAN objective, is an exponential moving average (EMA) of the generator's weights, a lightweight technique with a long history in GAN training. Yazıcı et al. [10] show theoretically and empirically that averaging generator weights over training acts as an implicit variance-reduction mechanism against the oscillatory dynamics of adversarial optimisation, improving both sample quality and training stability without any architectural change. Karras et al. [9], in StyleGAN2, adopt exactly this technique (an EMA-tracked copy of generator weights used at inference time) as a standard component of their state-of-the-art image generation pipeline, corroborating its practical value at scale. This paper's EMA fine-tuning stage is a direct, small-scale application of the same idea, and Part A of this assignment independently found EMA to be the single most effective technique among a wider set of ablations, motivating its reuse here as a candidate fix for any ACGAN trade-off.

2.4 Evaluation Metrics for Generative Models

Evaluating GAN sample quality requires metrics beyond simple visual inspection. The Inception Score [5] measures both the confidence and diversity of an Inception-v3 classifier's predictions on generated images, using a fixed pretrained classifier as a proxy for realism, but is sensitive to mode dropping within a class and does not compare directly against real-data statistics. Heusel et al. [4] address this with the Fréchet Inception Distance (FID), which compares the mean and covariance of Inception-v3 feature activations between real and generated image sets using the Fréchet distance between the two assumed-Gaussian distributions; a lower FID indicates the generated feature distribution is closer to the real one, and FID is the primary metric reported in this paper. Bińkowski et al. [8] propose the Kernel Inception Distance (KID) as an unbiased, lower-variance alternative that instead computes a polynomial-kernel Maximum Mean Discrepancy (MMD) between real and generated Inception features, without assuming Gaussianity. This paper reports KID alongside FID as a cross-check, computed via the same protocol as this study's ten-subset MMD estimator. Kynkäänniemi et al. [6] and Naeem et al. [7] further decompose the single-number FID/KID summaries into separate fidelity and diversity axes, specifically precision/recall and density/coverage, respectively, which Part A of this assignment used but which Part C does not directly recompute. Instead, Part C relies on FID/KID and an independent classifier's class-conditioning accuracy as its two complementary axes (distributional realism versus label correctness).

3. Dataset, Methodology and Experiment Design

3.1 Dataset

CIFAR-10 is used throughout, comprising 60,000 32×32 RGB images evenly split across ten mutually exclusive object classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). Pixel values are rescaled to the range [−1, 1] to match the generator's final tanh activation. Building directly on Part A's per-class error analysis, this paper designates two four-class groups for stratified evaluation:

- **Hard classes** (visually overlapping animals, historically difficult in Part A): cat, dog, bird, horse.
- **Structured/easier classes** (rigid, mechanically regular shapes with more consistent backgrounds): airplane, automobile, ship, truck.

3.2 Model Architectures

Both models share an identical generator, differing only in the discriminator. All parameter counts below are taken directly from the executed notebook's model summaries.

Generator (shared, 1,425,735 parameters). The generator takes a 128-dimensional latent noise vector and a class label. The label is passed through an Embedding(10, 50) layer, projected with a small dense layer, and reshaped to a single 4×4 feature map; the noise vector is separately projected through a dense layer to a 4×4×255 feature volume, batch-normalised, and LeakyReLU(0.2)-activated. The two paths are concatenated along the channel axis to form the initial 4×4×256 seed. Three upsampling stages follow, each consisting of a strided Conv2DTranspose (channel widths 128 → 64 → 32) with batch normalisation and LeakyReLU(0.2), followed by a same-resolution Conv2D refinement layer with its own batch normalisation and LeakyReLU(0.2). A final Conv2D with a 5×5 kernel and tanh activation produces the 32×32×3 RGB output.

Baseline discriminator (1,491,317 parameters). The class label is embedded and reshaped into a single-channel 32×32 map, concatenated with the input image along the channel axis before the convolutional backbone. The shared backbone consists of three blocks of Conv2D (4×4 kernel, stride 2) + LeakyReLU(0.2), each followed by a same-resolution Conv2D refinement + LeakyReLU(0.2) and Dropout(0.3), with channel widths 64 → 128 → 256. The flattened output feeds a single Dense(1) real/fake logit — the discriminator's only task is the adversarial decision, and it can only judge label consistency implicitly, through whatever correlation exists between the concatenated label map and the image.

ACGAN discriminator (1,478,539 parameters). The label is not fed into the discriminator at all. The same convolutional backbone (64 → 128 → 256 channels, identical structure to the baseline) processes the raw image alone, but the flattened features are split into two heads: a Dense(1) real/fake logit (the adversarial head) and a Dense(10) class-logits head (the auxiliary classifier). This dual-head design is the sole architectural difference between the two models and is the mechanism under test in this paper.

3.3 Loss Functions

The baseline model uses one-sided label smoothing (real target 0.9, fake target 0) with standard binary cross-entropy (BCE):

$$L_D(base) = BCE(\ D(x,\ y),\ 0.9\ ) + BCE(\ D(G(z,\ y),\ y),\ 0\ )$$
$$L_G(base) = BCE(\ D(G(z,\ y),\ y),\ 1\ )$$

where x is a real image, y its label, z a noise vector, and D(·, y) denotes the discriminator conditioned on the concatenated label map.

The ACGAN model adds a sparse categorical cross-entropy (SCE) classification loss, weighted by λ = 1.0 (LAMBDA_CLASS), computed on both real and fake images and applied to both the discriminator and generator objectives:

$$L_D(ACGAN) = [\ BCE(D_adv(x),\ 0.9) + BCE(D_adv(G(z,y)),\ 0)\ ]\ +\ \lambda\cdot[\ SCE(D_cls(x),\ y) + SCE(D_cls(G(z,y)),\ y)\ ]$$
$$L_G(ACGAN) = BCE(D_adv(G(z,y)),\ 1)\ +\ \lambda\cdot SCE(D_cls(G(z,y)),\ y)$$

Here D_adv and D_cls denote the two discriminator heads, and y for fake images is sampled uniformly at random over the ten classes each batch, forcing the generator to be able to produce every class on demand rather than only classes seen paired with a given noise region.

3.4 Training Configuration

Both models were trained for 80 epochs with a batch size of 64, Adam optimisers (learning rate = 2×10^{-4}, β1 = 0.5) for both generator and discriminator, a fixed random seed of 42 for reproducibility, and a 128-dimensional latent space. Steady-state training throughput was approximately 12.2–12.3 seconds per epoch for both models once the TensorFlow graph had warmed up, indicating the auxiliary classification head added negligible computational overhead relative to the adversarial-only baseline.

3.5 EMA Fine-Tuning Stage

Following the 80-epoch ACGAN run, an exponential moving average (EMA) shadow generator was initialised by copying the trained ACGAN generator's weights, then fine-tuned for a further 20 epochs at a reduced learning rate of 1×10^{-4}, continuing the same ACGAN-style adversarial + classification training. After every batch, the shadow generator's weights were updated as $\theta_{ema} \leftarrow 0.999\cdot\theta_{ema} + 0.001\cdot\theta_{live}$ (decay = 0.999), following the averaging principle described by Yazıcı et al. [10] and used in StyleGAN2 [9]. Only the EMA shadow generator, not the underlying continuously-updated live generator, is evaluated and reported as "ACGAN + EMA" in Section 4. Fine-tuning epochs ran at approximately 13.4–13.9 seconds each.

3.6 Evaluation Protocol

Distributional metrics (FID, KID). A frozen, ImageNet-pretrained InceptionV3 network (input resized to 75×75, global average pooling, classification head removed, non-trainable) extracts feature vectors for a fixed set of 2,000 real CIFAR-10 test images and for 2,000 class-balanced generated images (200 per class, generated with seeded, stateless noise for reproducibility). The same 2,000 real-image feature set is reused for all three evaluated models, ensuring a fair, identical reference distribution. FID is computed via the standard Fréchet distance between the Gaussian-approximated real and fake feature distributions (with a small $\varepsilon = 10^{-6}$ covariance regulariser for numerical stability in the matrix square root). KID is computed as an unbiased polynomial-kernel MMD^2 estimate, averaged over ten random subsets of 500 samples each.

Class-conditioning accuracy. Because grading a model's class-conditioning using its own auxiliary classifier head would be circular for the ACGAN, an independent evaluator CNN, trained only on real CIFAR-10 images and never exposed to any generated image, is used instead.This evaluator (two Conv2D(32) + BatchNorm + MaxPool + Dropout(0.25) blocks, two Conv2D(64) + BatchNorm + MaxPool + Dropout(0.30) blocks, a Conv2D(128) + BatchNorm + GlobalAveragePooling block, and a Dense(128) + Dropout(0.30) + Dense(10, softmax) head, trained with Adam at learning rate = 1×10^{-3} and early stopping) reached 78.24% test accuracy on real, held-out CIFAR-10 images which is a credible, independent basis for judging class identity. Class-conditioning accuracy for each generative model is then the fraction of 200 generated images per class (2,000 total, seeded per class) that this independent evaluator classifies as the requested label.

Per-class FID. For finer-grained diagnosis, FID is additionally computed separately for each of the ten classes, using 1,000 real and 1,000 generated images per class and the same InceptionV3 feature space and Fréchet-distance formula as the global metric. Hard-class-group and structured-class-group means are the simple averages of the four constituent per-class FID values in each group.

4. Results

4.1 Overall Comparison

Table 1 reports the three headline metrics for all three evaluated models, exactly as computed in the executed notebook.

Table 1. *Overall metrics: baseline cDCGAN vs. ACGAN vs. ACGAN+EMA.*

Model	FID ↓	KID ↓	Class-cond. acc. ↑	Hard-class mean FID ↓	Structured-class mean FID ↓
Baseline cDCGAN	258.9066	0.0868	63.15%	543.6195	234.1592
ACGAN	267.1537	0.0948	71.70%	548.5044	251.9535
ACGAN + EMA	248.7752	0.0799	76.25%	491.8118	229.4470

Moving from the baseline to plain ACGAN, class-conditioning accuracy rises by 8.55 percentage points (63.15% → 71.70%), supporting H1. At the same time, FID rises (worsens) from 258.91 to 267.15, and KID rises from 0.0868 to 0.0948. This represents a small but consistent worsening in both distributional-quality metrics, supporting the trade-off anticipated in H2. The hard-class mean FID is essentially flat (543.62 → 548.50, a 0.9% relative change), meaning ACGAN did not deliver a net improvement for the hard-class group, despite the large lift in labelling accuracy. The per-class breakdown below explains why.

Adding EMA fine-tuning on top of ACGAN improves every single metric simultaneously relative to plain ACGAN: FID falls to 248.78 (better than even the baseline's 258.91), KID falls to 0.0799 (better than the baseline's 0.0868), class-conditioning accuracy climbs further to 76.25%, and both the hard-class mean FID (491.81) and structured-class mean FID (229.45) become the best results of all three models. The ACGAN + EMA continuation therefore produced the strongest observed balance of distributional quality and class-conditioning accuracy. However, because it also included 20 additional fine-tuning epochs at a lower learning rate, the improvement cannot be attributed to EMA alone.

4.2 Per-Class Breakdown

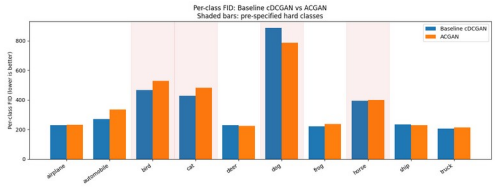
Table 2 lists per-class FID for all ten CIFAR-10 classes across the three models; hard classes (cat, dog, bird, horse) are marked with an asterisk.

Table 2. *Per-class FID by model (asterisk = hard class).*

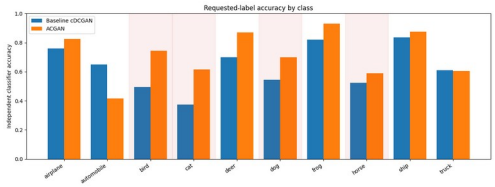
Class	Baseline cDCGAN	ACGAN	ACGAN + EMA
airplane	227.80	230.96	221.81
automobile	269.30	333.82	289.88
bird *	466.49	529.27	485.49
cat *	427.79	481.84	417.30
deer	229.25	224.39	216.42
dog *	885.76	784.96	690.68
frog	221.74	235.91	236.23
horse *	394.45	397.95	373.78
ship	234.20	228.54	218.75
truck	205.33	214.49	187.34

The picture within the hard-class group is asymmetric. Dog which is the single worst class for every model and the class Part A also flagged as hardest however it improves substantially under ACGAN (885.76 → 784.96, a 100.8-point drop) and improves further under EMA (→ 690.68). Horse worsens marginally under ACGAN (394.45 → 397.95) before improving to 373.78 under EMA. Bird and cat, however, both worsen under plain ACGAN relative to the baseline (bird: 466.49 → 529.27; cat: 427.79 → 481.84), and while EMA recovers most of this regression (bird → 485.49; cat → 417.30, now better than baseline), plain ACGAN alone leaves bird and cat worse off than the simpler baseline. Among the structured/easier classes, automobile shows the largest single regression under ACGAN (269.30 → 333.82), which EMA only partially reverses (→ 289.88, still above baseline).

Figure 1 visualises the per-class FID and requested-label accuracy comparisons between baseline and ACGAN directly (side by side below), making the hard-class asymmetry easy to see immediately.



(a) Per-class FID



(b) Per-class requested-label accuracy

Figure 1. *(a) Per-class FID, baseline cDCGAN vs. ACGAN — hard classes (cat, dog, bird, horse) shaded; dog improves substantially under ACGAN while bird and cat worsen. (b) Per-class class-conditioning accuracy (independent evaluator), baseline cDCGAN vs. ACGAN.*

4.3 Qualitative Samples

Figure 2 shows class-conditioned sample grids generated by each of the three models at the end of training (side by side below for direct visual comparison), each row corresponding to one CIFAR-10 class, produced directly from the notebook's plotting cells with no post-processing.



(a) Baseline cDCGAN *(b) ACGAN* *(c) ACGAN + EMA*

Figure 2. *Class-conditioned sample grids (rows = CIFAR-10 classes, 80 training epochs unless noted). (a) Baseline cDCGAN. (b) ACGAN. (c) ACGAN + EMA, after the 20-epoch EMA fine-tuning stage.*

Figure 3 shows the generator and discriminator adversarial loss curves recorded during the 20-epoch EMA fine-tuning stage, included to confirm that fine-tuning proceeded stably (no collapse or runaway divergence) rather than to claim any quantitative loss-based conclusion, since GAN losses are not reliable proxies for sample quality on their own.

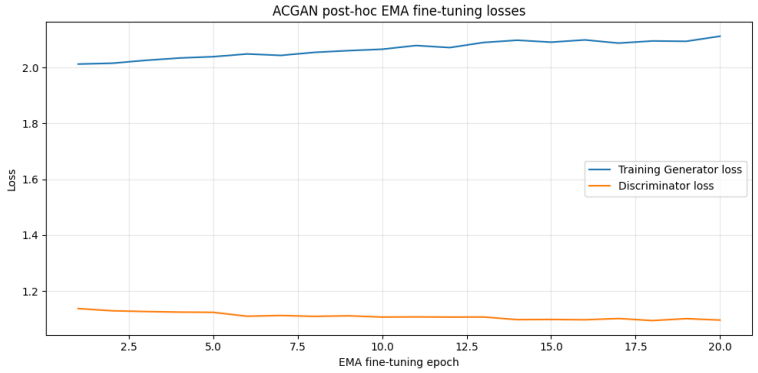


Figure 3. *Generator/discriminator adversarial loss during the 20-epoch EMA fine-tuning stage.*

5. Discussion

5.1 Interpreting the Trade-off

The results give a clear, if nuanced, answer to the research question. Adding an auxiliary classification objective does improve class-conditional fidelity in the sense that matters most directly. An independent classifier is substantially more likely to agree that a generated image belongs to its requested class (63.15% → 71.70%), which supports H1 unambiguously. But "class-conditional fidelity" is not single-dimensional: it also worsened the models' overall statistical resemblance to real CIFAR-10 images (FID and KID both rose), and its effect on the specific hard-class group this paper set out to help was a wash rather than a win, because two of the four hard classes (bird, cat) moved in the wrong direction. This directly supports H2, and the same asymmetric pattern (large gains for the numerically worst class, regressions for visually similar mid-difficulty classes) is consistent with the asymmetric behaviour predicted by the ACGAN diversity-collapse literature.

Gong et al.'s theoretical account [14] offers a plausible mechanism for the bird/cat regression specifically. The auxiliary objective rewards easily classifiable samples, which may encourage the Generator toward highly separable but narrower regions of a class distribution. This could improve requested-label accuracy while worsening distributional coverage for overlapping classes. This effect would be expected to fall hardest on classes whose real supports substantially overlap (as bird and cat overlap with other small four-legged or feathered classes far more than dog, whose silhouette and features are comparatively distinctive), worsening the match to the full real-data distribution captured by FID even as the nominal class label becomes easier for a downstream classifier to confirm. The related concern raised by Hou et al. [15], that an auxiliary classification signal can become progressively less informative as the generator's distribution shifts during training, is also consistent with why dog (the class with the most room

for improvement and, correspondingly, the strongest classification gradient signal throughout training) benefited most, while already-closer classes such as bird and cat received a comparatively noisier or less well-calibrated classification gradient.

5.2 Interpreting the EMA Continuation Stage

The result that the EMA continuation improves every metric, rather than merely trading fidelity for conditioning accuracy along the same frontier, is the most interesting finding of this study, and it echoes Part A's independent discovery that EMA was the single most effective technique in that assignment's own ablation sequence. It is important to be precise about attribution here: the continuation stage changed three things at once. It applied EMA weight averaging, added 20 further training epochs and lowered the learning rate. Therefore, the observed gain cannot be credited to weight averaging alone, and an ablation isolating EMA from the additional low-learning-rate epochs would be required to separate these factors. EMA itself does not change the loss function or the conditioning mechanism; it only averages weights over a stabilisation window. Yazıcı et al.'s argument [10] that adversarial training oscillates around (rather than converging smoothly to) a local equilibrium, and that weight averaging suppresses this oscillation without changing what equilibrium is being approached, explains why EMA can improve both axes here: if plain ACGAN's bird/cat regressions are partly attributable to noisy, oscillatory late-training dynamics compounding the auxiliary classifier's separability pressure, then averaging over the last several epochs of training would be expected to reduce exactly that noise, recovering some of the lost fidelity without undoing the classification gradient's conditioning benefit (which is present throughout the averaging window, not just at one noisy final checkpoint). This is consistent with the StyleGAN2 authors' [9] adoption of EMA as a default, architecture-agnostic component rather than a fix for any one specific instability.

5.3 Relation to Part A and to the Broader Literature

This paper's baseline cDCGAN (FID 258.91, class-conditioning accuracy 63.15%) is numerically worse on both axes than Part A's fully-tuned final model (FID 103.86, class-conditioning accuracy 80.1%), which is expected: Part A's baseline used a smaller/shallower architecture trained for fewer epochs before capacity increases and a longer 150-epoch EMA schedule were applied, whereas this paper deliberately holds capacity and epoch budget constant between its two models to isolate the conditioning-mechanism variable in a controlled, single-factor comparison. The two papers are complementary rather than contradictory: Part A shows that capacity and training-length/EMA improvements are effective independently of the conditioning mechanism, while Part C shows that the conditioning mechanism itself is a separate, additional lever and that its effect, unlike a straightforward capacity increase, is genuinely two-sided rather than uniformly beneficial.That the hard classes identified by Part A's capacity-focused study (cat, dog, bird and horse) do not respond uniformly to Part C's conditioning-focused intervention is itself informative. It suggests that "hardness" for CIFAR-10 animal classes is not a single underlying cause that any one intervention resolves, but a mix of at least two distinct issues: insufficient model capacity or training, which was addressed by Part A, and insufficiently separable real-data classes interacting poorly with an auxiliary classifier, which is the mechanism that Section 5.1 attributes to bird and cat.

The projection discriminator of Miyato and Koyama [12], which conditions the discriminator without any explicit classification loss, is a natural next comparison this paper did not implement but which the results motivate: if the bird/cat regression is specifically caused by the classification loss term rather than by removing the label from the discriminator's direct input, a projection discriminator should retain ACGAN's conditioning benefit without its fidelity cost, since it conditions the discriminator by inner-product rather than by an explicit, separability-seeking classification objective.

5.4 Limitations and Threats to Validity

Several limitations qualify these conclusions. First, all results come from a single random seed (42) and a single training run per model; no confidence intervals or repeated-run variance estimates were computed, so small differences (e.g. horse's FID moving from 394.45 to 397.95 under ACGAN) could plausibly fall within run-to-run noise rather than reflecting a genuine effect, though the larger, consistent differences (dog's ~100-point FID improvement, bird/cat's 50+ point regressions, the 8.55-point accuracy gain) are unlikely to be noise alone given their magnitude relative to the smaller structured-class movements. Second, FID and KID are computed using an ImageNet-pretrained InceptionV3 feature extractor operating on images resized from 32×32 to 75×75; this is a standard but imperfect proxy, since the feature extractor was never trained on CIFAR-10-scale images, and absolute FID values in the 200–900 range reported here are much higher than typical FID values reported in higher-resolution GAN literature, reflecting both CIFAR-10's low native resolution and this evaluation protocol's specific implementation rather than being directly comparable to FID values from other papers using different feature extractors or resolutions. Third, both models were trained for a comparatively short 80 epochs (plus a 20-epoch EMA fine-tune) relative to the hundreds of epochs typical in published ACGAN or StyleGAN2 experiments, so it remains possible that the bird/cat regression is partly a symptom of insufficient training time for the auxiliary classifier's gradient to stabilise, rather than a permanent property of the ACGAN objective. Part A's own finding that its baseline model improved substantially between 80 and 150 epochs suggests that training length is a real, unaddressed confound

here. Finally, the independent evaluator achieved 78.24% accuracy on real test images, so its predictions introduce measurement noise and should be interpreted as an approximate indicator of conditional fidelity rather than an exact measure, for all three generative models.

6. Conclusion and Future Work

This paper set out to test whether an auxiliary classification objective can improve class-conditional fidelity for visually overlapping CIFAR-10 classes relative to a standard label-concatenation conditional DCGAN, building directly on Part A's finding that cat, dog, bird, and horse are disproportionately hard classes for conditional GAN generation. Under matched architecture, hyperparameters, and training budget, ACGAN raised independent class-conditioning accuracy from 63.15% to 71.70% but slightly worsened FID (258.91 → 267.15) and KID (0.0868 → 0.0948), with an uneven hard-class effect: dog improved substantially while bird and cat worsened, leaving the hard-class-group mean FID essentially flat. This supports both hypotheses and reproduces, in a small-scale CIFAR-10 setting, the diversity/fidelity trade-off documented theoretically for ACGAN by Gong et al. [14] and Hou et al. [15]; with a single run per model, these results support rather than formally confirm a general claim. A subsequent 20-epoch EMA continuation then produced the strongest observed balance of distributional quality and class-conditioning accuracy, yielding a model better than both the baseline and plain ACGAN on FID (248.78), KID (0.0799), class-conditioning accuracy (76.25%), and hard-class mean FID (491.81). However, because this stage also included 20 additional fine-tuning epochs at a lower learning rate, the improvement cannot be attributed to EMA alone.

The main practical takeaway is that auxiliary classification and weight averaging address different failure modes and are complementary rather than substitutable: the classification loss supplies a conditioning signal that label-concatenation alone lacks, while the EMA continuation was consistent with reduced training instability, although the contribution of EMA cannot be separated from the additional epochs and lower learning rate. Future work suggested directly by this paper's limitations and Section 5.3's discussion includes:

1. repeating the baseline-vs-ACGAN comparison across multiple random seeds to establish confidence intervals on the per-class FID differences;
2. extending training length to match Part A's 150-epoch budget to check whether the bird/cat regression persists or is a short-training artefact;
3. implementing a projection discriminator [12] as a third conditioning strategy to test whether it retains ACGAN's conditioning benefit without its fidelity cost, since it conditions without an explicit classification loss; and
4. combining the auxiliary classifier with a Twin Auxiliary Classifier [14] or Auxiliary Discriminative Classifier [15] correction directly, rather than relying on EMA as a post-hoc fix, to test whether these purpose-built corrections outperform the simpler EMA-based mitigation used here.

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